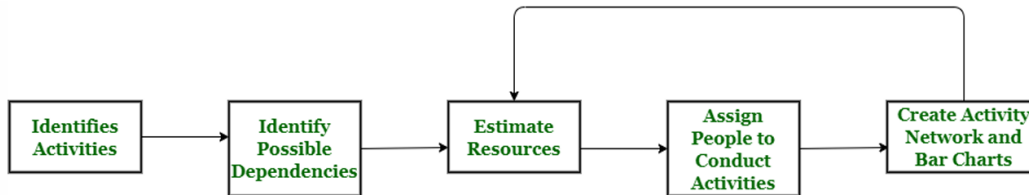


Chapter 3 Project Scheduling

Introduction

- Project is a **Problem Scheduled for Solution**.
- Scheduling Project Activities is a **Significant Part of Project Preparation**.
- It includes **determining which activities(tasks) will be completed and when**.

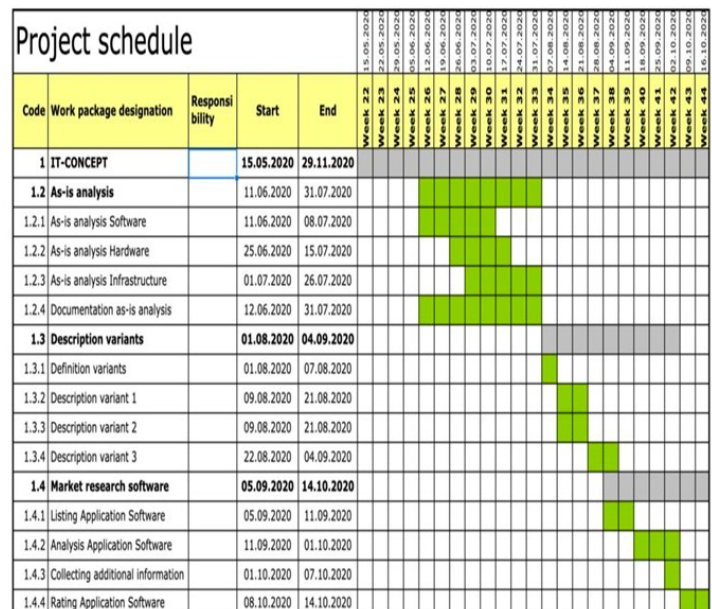


A Software Project Manager needs to do the following:

1. identify all the activities needed to complete the Project.
2. Break-down (Divide) Complex task into Smaller activities.
3. Determine the Dependency among different activities.
4. Establish the most likely estimates for the time durations necessary to Complete the activities.
5. Allocate Resources to activities.
6. Plan the Starting and Ending dates for various activities.
7. Determine the Critical Path.

Gantt Chart

- A Gantt chart is a **project management tool assisting in the planning and scheduling of projects of all sizes; they are particularly useful for visualizing projects**.
- A Gantt chart is defined **as a graphical representation of activity against time; it helps project professionals monitor progress**.

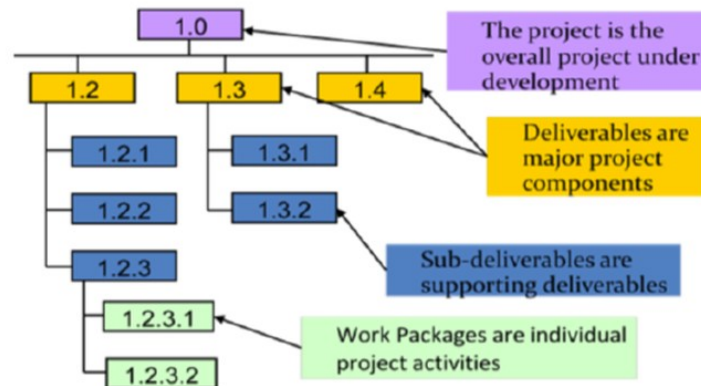


- Show task durations.
- Show time overlap.
- Show slack time in duration.

❑ Work Breakdown Structure (WBS)

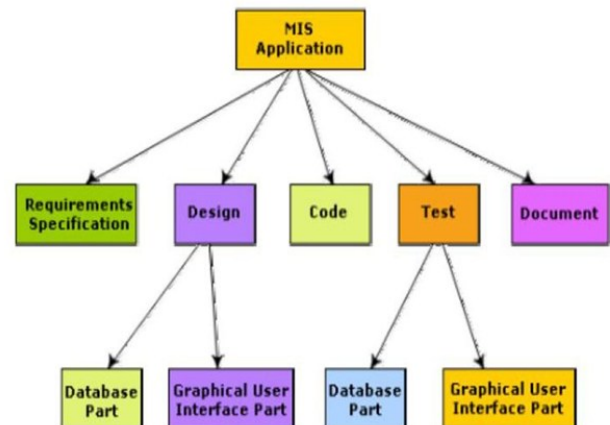
- A work-breakdown structure (WBS) in project management and systems engineering is a deliverable-oriented breakdown of a project into smaller components.
- A work breakdown structure is a key project management element that organizes the team's work into manageable sections.

Work Breakdown Structure Deliverables



Work Breakdown Structure of an MIS

- Show task dependencies.
- Show slack time in boxes.
- Do not show time overlap, but show parallelism.

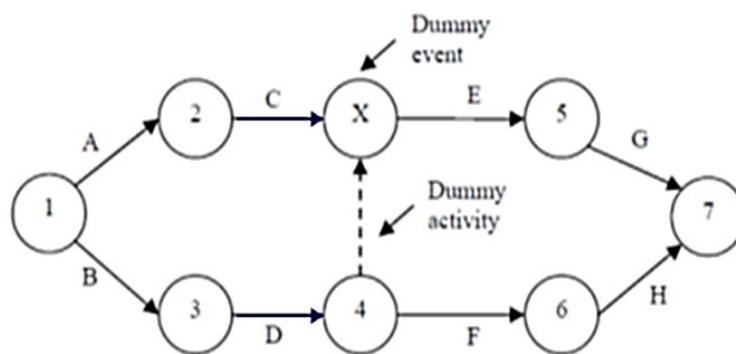


Example of WBS for a Project

Breakdown	Description	WBS	Code
IT Installation Project			1.0
Deliverable 1	Match IT to organizational tasks and problems		1.1
WP 1	Conduct problem analysis	1.1.1	
WP 2	Develop information on IT technology	1.1.2	
Deliverable 2	Identify IT user needs		1.2
WP 1	Interview potential users	1.2.1	
WP 2	Develop presentation of IT benefits	1.2.2	
WP 3	Gain user "buy-in" to system	1.2.3	
Deliverable 3	Prepare informal proposal		1.3
WP 1	Develop cost/benefit information	1.3.1	
WP 2	Gain top management support	1.3.2	
Deliverable 4	Seek and hire IT consultant		1.4
WP 1	Delegate members as search committee	1.4.1	
WP 2	Develop selection criteria	1.4.2	
WP 3	Interview and select consultant	1.4.3	
Deliverable 5	Seek staff and departmental support for IT		1.5
Deliverable 6	Identify the appropriate location for IT		1.6
WP 1	Consult with physical plant engineers	1.6.1	
WP 2	Identify possible alternative sites	1.6.2	
WP 3	Secure site approval	1.6.3	
Deliverable 7	Prepare a formal proposal for IT introduction		1.7
Deliverable 8	Solicit RFPs from vendors		1.8
WP 1	Develop criteria for decision	1.8.1	
WP 2	Contact appropriate vendors	1.8.2	
WP 3	Select winner(s) and inform losers	1.8.3	
Deliverable 9	Conduct a pilot project (or series of projects)		1.9
Deliverable 10	Enter a contract for purchase		1.10
Deliverable 11	Adopt and use IT technology		1.11
WP 1	Initiate employee training sessions	1.11.1	
WP 2	Develop monitoring system for technical problems	1.11.2	

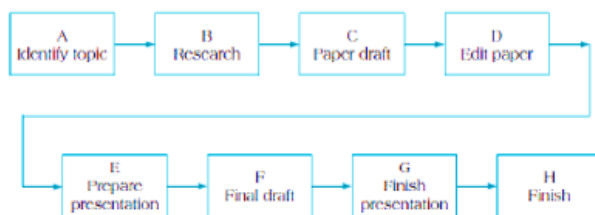
❑ Developing Network Diagram

- Assuming the Students are Preparing a Seminar, to ensure the best use of your time and availability, it would be necessary to determine a Reasonable (Logical) Sequence.
- Preceding activities** (are those that must occur before others can be done).
- Subsequent or Successor activity** (are those that must follow others).
- For example**, it would be necessary to first Identify the term paper topic before beginning to Conduct research on it.
- Therefore, Activity1, Identify topic, is a Preceding activity; and Activity2, Research topic, is referred to as Successor activity.
- Once you have Identified a Reasonable Sequential Logic for the Network, you can Construct a Network Diagram.
- Network Diagram is a Schematic display of the Project's Sequential Activities and the Logical Relationships between them.

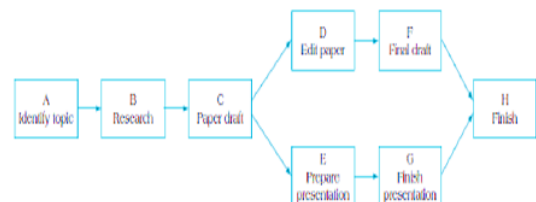


Alternative Activity Networks for Term Paper Assignment

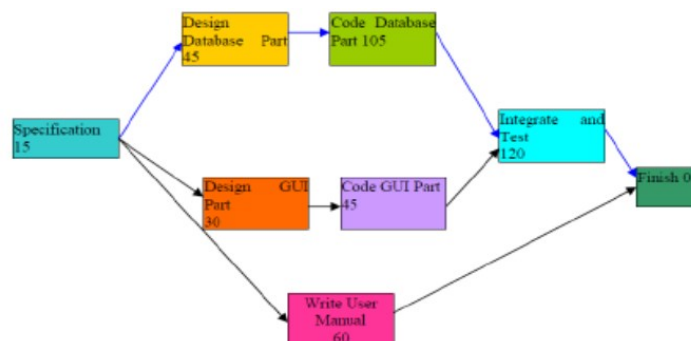
Option A: Serial Sequential Logic



Option B: Non-Serial Sequential Logic



Activity Network Representation of the MIS Problem



❑ Network

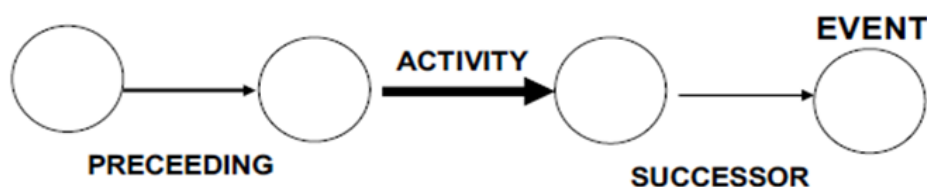
- Network - Graphical Portrayal of activities and events.
- Shows Dependency Relationships between activities or tasks in a Project.
- Clearly shows tasks that must Precede (Precedence) or Follow (Succeeding) other tasks. in a logical manner (Reasonable Sequence).
- Clear Representation of Plan – a Powerful tool for Planning & Controlling Project.
- Combination of all Project activities and the Events

❑ Rules for Constructing a Network Diagram

1. Some determination of activity Precedence Ordering must be done before Creating the Network. That is, all activities must be logically linked to each other - those that precede others, as well as Successor Activities (those that must follow others).
2. Network diagrams usually flow from left to right.
3. An activity Cannot begin until all Preceding connected activities have been Completed.
4. Arrows (Arcs) on Networks indicate Precedence and Logical Flow.
5. Each activity should have a Unique Identifier associated with it (number, letter, code, etc.). For simplicity, these identifiers should occur in Ascending Order; each one should be larger than the identifiers of Preceding activities.
6. Looping, or Recycling through activities, is Not Permitted.
7. Although not required, it is common to Start a Project from a Single beginning node, even in the case when multiple start points are possible. A single node point also is typically used as a project End indicator.

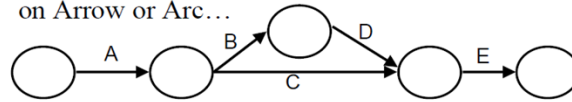
Definition of Terms in a Network

- **Activity:** any Portions of Project (tasks) which required by Project, uses up resource, and consumes time – may involve (labor, paperwork, contractual negotiations, machinery operations).
- **Activity on Arrow(AOA)** showed as arrow(AON)– Activity on Node.
- **Event:** Starting or Ending Points of one or more activities, instantaneous point in time, also called 'Nodes'

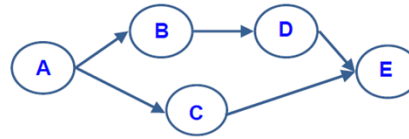


AOA Versus AON

The same mini-project is shown with Activities on Arrow or Arc...



... and Activities on Node.



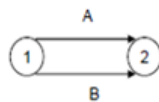
Rules for Constructing a Network Diagram

Construction of Network should be based on logical or technical dependencies among activities.

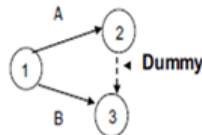
1. Each activity is represented by One and Only One Arrow (Arc) in the Network = (Unique Node);
2. No two activities can be identified by the same head and tail events (a dummy activity is introduced in such situations);

Example of the Use of Dummy Activity

Network concurrent activities

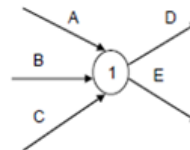


WRONG !!!

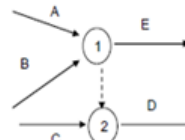


RIGHT ✓

Activity C not required for E



WRONG !!!



RIGHT ✓

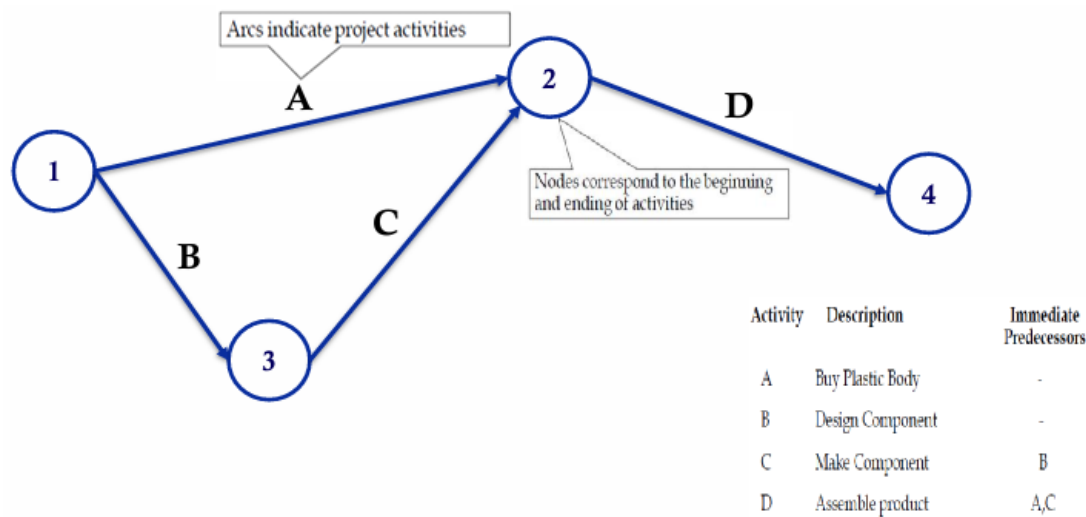
➤ Example1 -A Simple Network

Consider the list of four activities for Making a Simple Product:

Immediate Predecessors for a particular activity are the activities that, when completed, enable the start of the activity in question.

Activity	Description	Immediate Predecessors
A	Buy Plastic Body	-
B	Design Component	-
C	Make Component	B
D	Assemble product	A,C

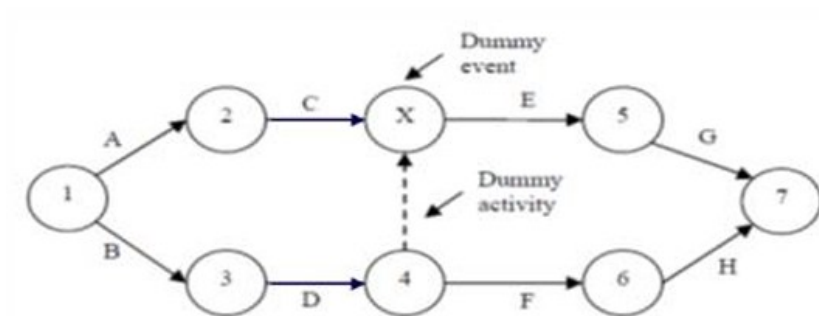
Answer



➤ **Example2 Develop the network for a Project with the table below:**

Activity	Immediate Predecessors
A	-
B	-
C	A
D	B
E	C, D
F	D
G	E
H	F

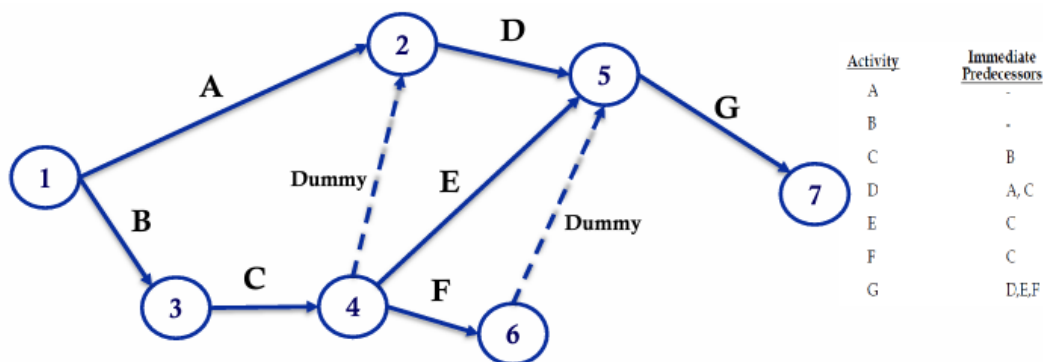
Answer



➤ **Example 3 Develop the network for a Project with the table below:**

<u>Activity</u>	<u>Immediate Predecessors</u>
A	-
B	-
C	B
D	A, C
E	C
F	C
G	D,E,F

Answer

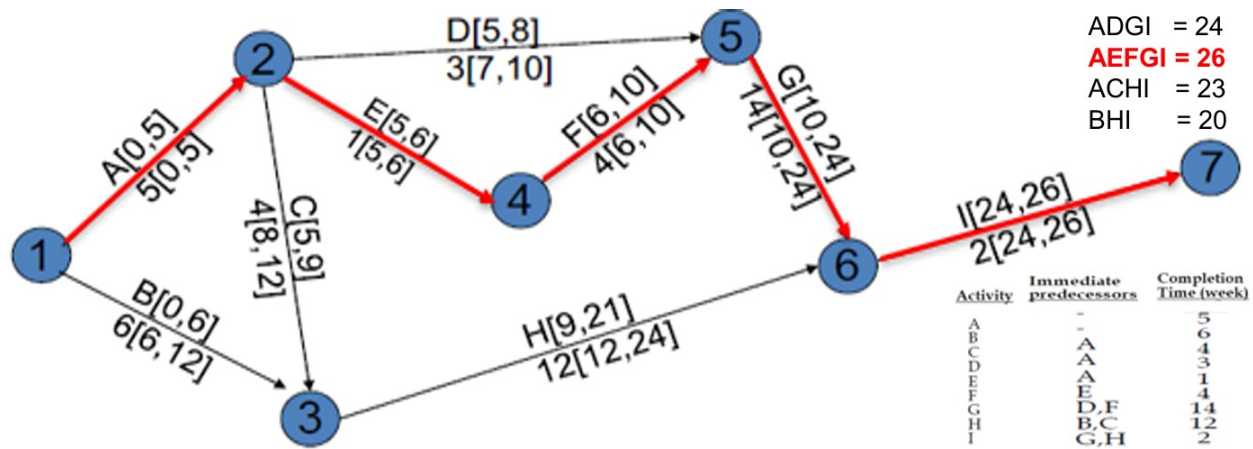


Note how the Network correctly identifies D, E, and F as the Immediate Predecessors for G.

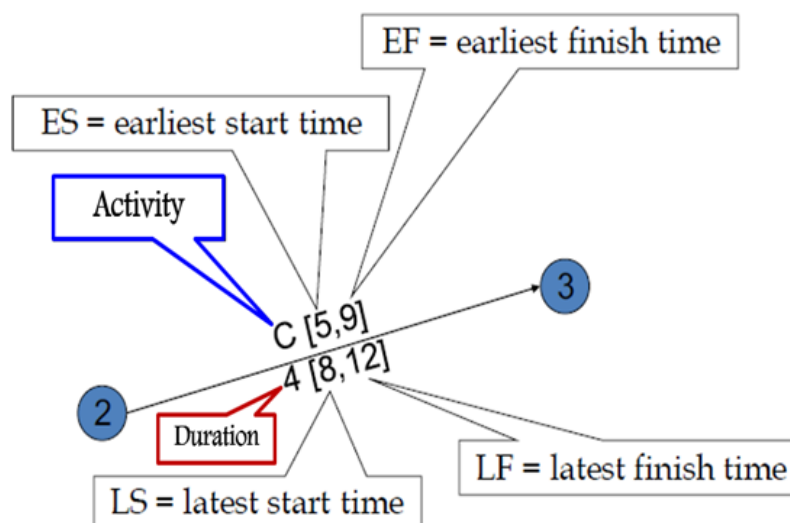
Dummy activity is used to identify Precedence Relationships correctly and to Eliminate Possible Confusion of two or more activities having the same starting and ending nodes.

Dummy activities have No Resources (time, labor, machinery, etc.) – Purpose is to PRESERVE LOGIC of the Network.

❑ Construct the CPM (Identify Activity Float)



- Following the Completion of the **Project Network Plan** and the **Understanding of activity times**, we think about **How Long the Project will take to be Finished** and **When the tasks will be Planned**. This can be addressed by determining the **Network's Critical Path Method (CPM)**.
- For this, we need an **Arrow Graph** and the **Timeframe** of all the activities.
- These Calculations **integrate a Forward (ES, EF) and a Backward (LS, LF) Crossing in the Network**.
- The Forward Crossing** **estimations harvest the First Start and the First Finish durations** for each activity.
- The Backward Crossing** **estimations render the Latest allowable Start and the Latest Finish durations** for every activity.
- With the aid of the Examples given, we will **illustrate how to calculate the Earliest Start, Earliest Finish, Latest Start, and Latest Finish times** of different project activities.



Scheduling with activity time

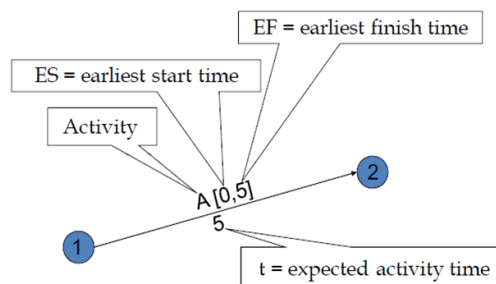
<u>Activity</u>	<u>Immediate predecessors</u>	<u>Completion Time (week)</u>
A	-	5
B	-	6
C	A	4
D	A	3
E	A	1
F	E	4
G	D,F	14
H	B,C	12
I	G,H	2
Total ...		51

- This information indicates that the total time required to complete activities is 51 weeks. However, we can see from the network that Several of the activities can be Conducted simultaneously (A and B, for example).

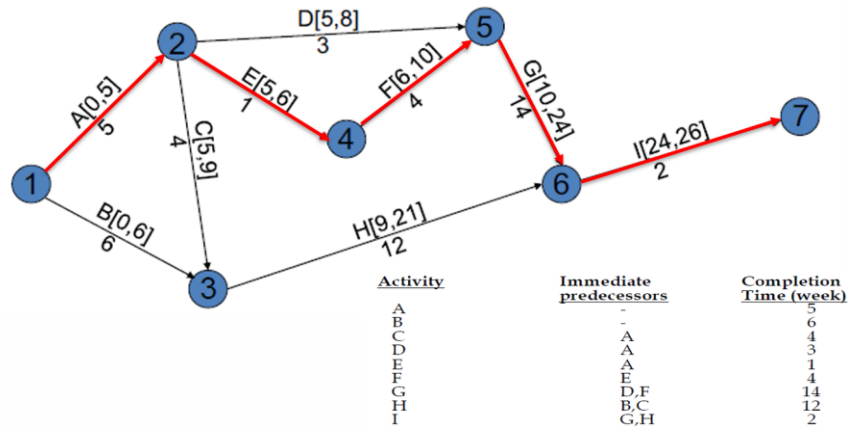
➤ Earliest Start & Earliest Finish Time

- We are interested in the Longest Path through the Network, i.e., the Critical Path.
- Starting at the Network's Origin (Node1) and using a starting time of 0, we Compute an Earliest Start (ES) and Earliest Finish (EF) time for each activity in the network.
- The Expression $EF = ES + t$ can be used to find the earliest finish time for a given activity.
- For Example, for activity A, $ES=0$ and $t=5$; thus, the earliest finish time for activity A is $EF = 0 + 5 = 5$

Time Arc with [ES & EF] time



Network with ES & EF time for all activities



➤ Earliest Start Time Rule:

- The Earliest Start time for an activity leaving a particular node is equal to the Largest of the Earliest Finish times for all activities entering the node.

➤ Latest Finish Time Rule:

- The Latest Finish time for an activity entering a particular node is equal to the Smallest of the Latest Start times for all activities leaving the node.

➤ Latest Start & Latest Finish Time

- To find Critical Path we need a Backward Pass calculation.
- Starting at the Completion Point (Node7) and using a latest finish time (LF) of 26 for activity I, we trace back through the network computing a Latest Start (LS) and Latest Finish time for each activity.
- The Expression $LS = LF - t$ can be used to calculate latest start time for each activity.
- For Example, for activity I, $LF=26$ and $t=2$; thus, the latest start time for activity I is $LS = 26 - 2 = 24$

➤ Slack or Free Time or Float

- Slack/Float refers to the amount of time an operation may be Postponed Without Changing the Project's Overall End Date. Slack for C= 3 weeks.
- For Example, Activity C can be postponed for up to 3 weeks (Starting somewhere between week 5 and 8).

Activity Schedule for our Example

Activity	Earliest Start (ES)	Latest Start (LS)	Earliest Finish (EF)	Latest Finish (LF)	Slack (LS-ES)	Critical Path
A	0	0	5	5	0	Yes
B	0	6	6	12	6	
C	5	8	9	12	3	
D	5	7	8	10	2	
E	5	5	6	6	0	Yes
F	6	6	10	10	0	Yes
G	10	10	24	24	0	Yes
H	9	12	21	24	3	
I	24	24	26	26	0	Yes

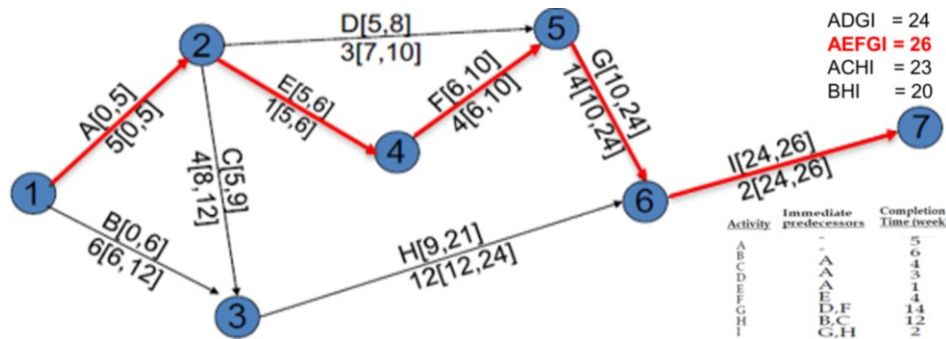
Critical Path identifies the minimum time to Complete Project

➤ Critical Path (CP)

What is the total time to Complete the Project (Critical Path)?

Activity	Immediate predecessors	Completion Time (week)
A	-	5
B	-	6
C	A	4
D	A	3
E	A	1
F	E	4
G	D, F	14
H	B, C	12
I	G, H	2
Total ...		26

- 26 Weeks if the individual activities are Completed on Schedule.



The End of Chapter 3 Part 1

لا تنسوا الدعاء لوالدي بالرحمة والمغفرة